MILITARY SPECIFICATION

TELESCOPE, ELBOW ASSEMBLY: 8211640

This Amendment forms a part of MIL-T-63443 {AR), dated 22 March 1982, and is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1, DRAWINGS, add:

"F7687113 Test Fixture, Aiming Circle
F7680631 Dioptometer
C7641866 Surface Quality Standards for Optical Elements (Scratch)
F8565556 Pressure Tester - Two Station"

PAGE 3

3.4.8, add new paragraph:

"3.4.8 Collimation. Collimation shall be in accordance with note 2 of drawing F8211640."

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TABLE I, add Characteristic 108:

"108.	Collimation	3.4.8	4.6.10"

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4.6.10, add new paragraph:

"4.6.10 Collimation. The collimation test for the requirement of 3.4.8 shall be conducted with the special testing equipment conforming to drawing 7687113, and in accordance with the set-up and operation instructions specified as follows:

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a. With collimator “A” in Supports "B" and “C”, remove wrench “D" and bushing “E” from plate “F”.

b. Plumb collimator “A” to a toolmaker’s square located on surface “N”.

c. Locate mirror “G” on pin “H” with front surface of mirror against plate "F” and held in place by clamp “I”.

d. By use of adjusting screws “J”, auto-collimate the horizontal reticle line of collimator “A”.

e. Remove mirror “G” and recheck plumb as noted under b above. If adjustments are required, such adjustments shall be made until requirements b and d above are met.

f. Insert mirror assembly “K” into locating ring “M” with front surface of mirror “G” against surface “O” and facing collimator “A”.

g“ Hold mirror assembly “K” with mirror “G” in a vertical position and secure with clamp “L”.

h. Check vertical reticle of collimator and ascertain if it is superimposed on mirror assembly “K”. If vertical reticle of collimator “A” is not superimposed on mirror assembly “K”, adjust collimator accordingly.

i. Recheck b through h inclusive until all requirements are met. Then remove mirror assembly from test equipment.

j. Replace wrench “D” and bushing “E” in plate “F”.

k. Back off wrench “D” and bushing “E” sufficiently to clear the objective end of the telescope.

1. Insert telescope into locating ring “M” with the objective end of the telescope pointed toward collimator “A” and secure with clamp “L”; do not tighten.

m. Insert bushing “E” into objective end of the telescope. When bushing “E” is firmly seated, tighten clamp “L”.

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n. Insert wrench “D” into objective cell of telescope, and while sighting through eyelens of telescope, reduce parallax to the specification tolerances.

0. While sighting through eyelens to telescope, plumb and collimate telescope reticle to collimator “A” reticle by means of the reticle adjusting screws of the telescope.”

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(Project 1290-A431)

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